

Madhumithra

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Professional Summary

Final-year B.Tech CSE (Information Security) student at VIT with strong skills in Java, DSA, OOP, and problem-solving. Experienced in developing AI-powered and full-stack applications using Python, Flask, Next.js, and Firebase, with a strong interest in software engineering and secure technology solutions.

Education

Vellore Institute of Technology, Vellore, B.Tech Computer Science and Engineering Aug 2023 – May 2027
(Information Security)

CGPA: 9.38 / 10.0

Government Girls Higher Secondary School, Pattukkottai Mar 2023
Higher Secondary Education – 95.83%

Projects

AI PrepZone – AI-Powered Smart Learning Platform [\[GitHub\]](#)

- Built an AI-powered exam preparation platform using Flask and the Gemini API for personalized quiz generation.
- Developed a dual-chatbot system: one chatbot retrieves previous years' question papers using Dialogflow, while the other uses the Gemini API to summarize and explain pasted questions.
- **Technologies:** Python, Flask, Gemini API, Dialogflow, Pandas, HTML, CSS, JavaScript

Collaborative Sheets – Real-Time Spreadsheet Collaboration [\[Live Demo\]](#)

- Built a full-stack real-time collaborative spreadsheet application inspired by Google Sheets using Next.js and Firebase.
- Implemented live multi-user editing, Firebase Authentication, real-time synchronization, custom formula calculations, cell formatting, and CSV export.
- **Technologies:** Next.js 14, React, TypeScript, Firebase Authentication, Firebase Realtime Database, Tailwind CSS

Patents

Published Indian Patent Application: System for Risk-Based Stateful Control of Artificial Intelligence Agent Interactions [\[Patent\]](#) 2026

- Publication No.: IN202641068105 A1.
- Designed a stateful, risk-based security framework integrating adaptive risk scoring, context management, and attack memory to protect AI agents and LLMs from advanced prompt-based attacks.

Certification

IBM Cybersecurity Analyst Professional Certificate [\[Certificate\]](#)

Completed IBM Cybersecurity Analyst training covering cybersecurity fundamentals, SOC, SIEM, SOAR, and incident management.

Technical Skills

Languages: Java

Core CS: DSA, OOP, DBMS, OS, CN

Web: HTML, CSS, React, Next.js

Databases: Firebase, SQL

Tools: Git, GitHub, VS Code

Achievements

VIT STARS Scholarship: Recipient of a 100% VIT STARS tuition scholarship for academic excellence.

Academic Excellence: School Topper in both 10th and 12th Grade Board Examinations.

Coding Practice: Solved 140+ DSA problems on Leetcode.